# Experiment Analysis

# Experiment Analyisis

- Hypothesis: Mac menubar is faster to access than Windows menubar
  - Design: between-subjects, randomized assignment of interface to subject

| Windows | Mac |
|---|---|
| 625 | 647 |
| 480 | 503 |
| 621 | 559 |
| 633 | 586 |

Each user used only one version of the menu bar

each user did one trial, clicking on the menu bar just once while we timed their speed of access. (only one trial is a very unreliable experiment design)

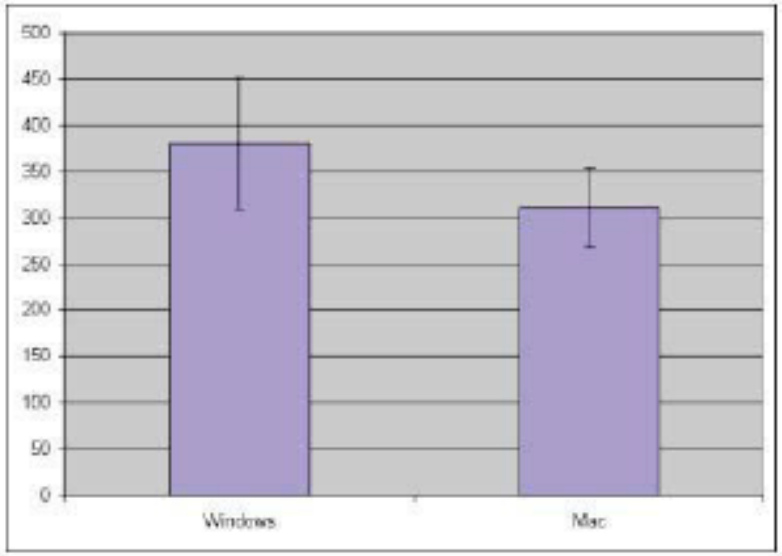
# Statistical Testing

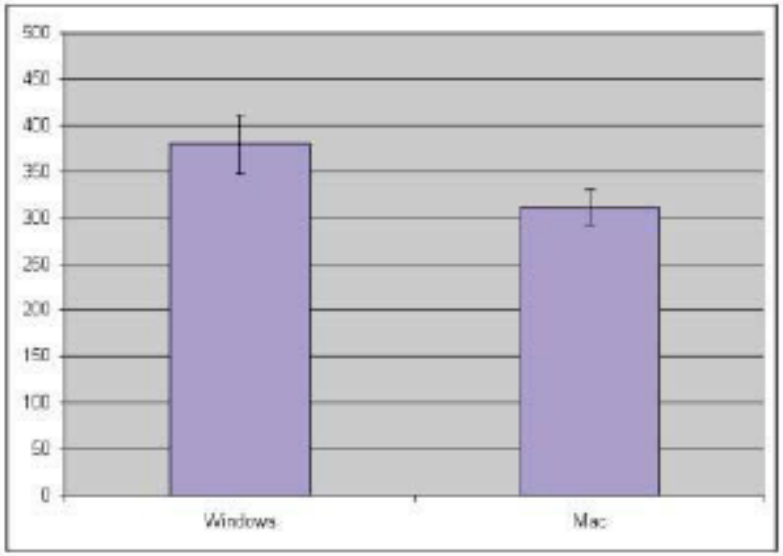
- Compute a statistic summarizing the experimental data

  mean(Win)
  mean(Mac)

- Apply a statistical test
  - *t* test: are two means different?
  - ANOVA (ANalysis Of VAriance): are three or more means different?
- Test produces a *p* value
  - *p* value = probability that the observed difference happened purely by chance
  - If $p < 0.05$, then we are 95% confident that there is a difference between Windows and Mac

# Standard Error of the Mean



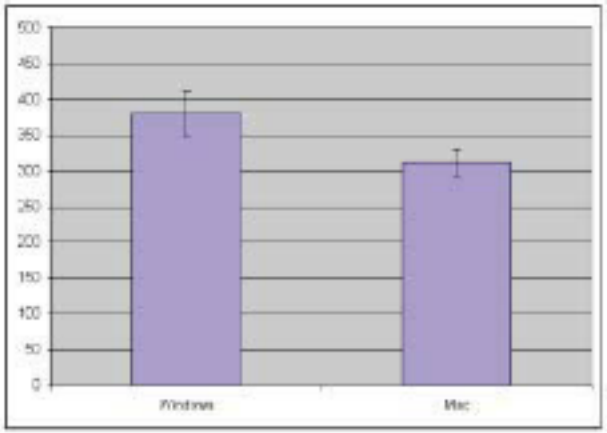




N = 4 :
Error bars overlap, so can't conclude anything

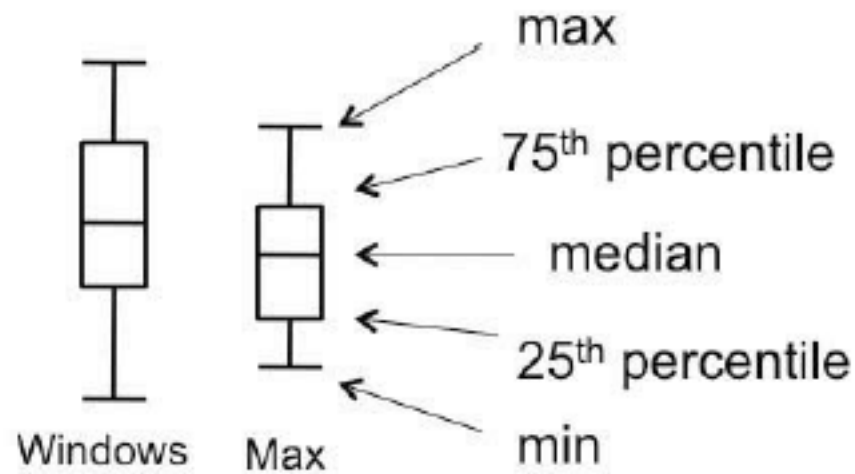
N=10:
Error bars are disjoint, so Windows may be different from Mac

stretching 1 standard error above and below the top of each bar

# Graphing Techniques




**Error bars**




**Tukey box plots**

- Pros
  - Easy to compute
  - Give a feel for your data
- Cons
  - Not a substitute for statistical testing

# Hypothesis Testing

- Our hypothesis: position of menubar matters
  - i.e., mean(Mac times) < mean(Windows times)
  - This is called the alternative hypothesis (also called H1)
- If we're wrong: position of menu bar makes no difference
  - i.e., mean(Mac) = mean(Win)
  - This is called the null hypothesis (H0)
- We can't really disprove the null hypothesis
  - Instead, we argue that the chance of seeing a difference **at least as extreme** as what we saw is very small if the null hypothesis is true

# Running a T Test

| Win | Mac |
|---|---|
| 625 | 647 |
| 480 | 503 |
| 621 | 559 |
| 633 | 586 |

| | *Windows* | *Mac* |
|---|---|---|
| Mean | 589.8 | 573.9 |
| Variance | 5368.5 | 3574.0 |
| Observations | 4 | 4 |
| Pooled Variance | 4471.2 | |
| Hypothesized Mean Difference | 0 | |
| df | 6 | |
| t Stat | 0.336 | |
| P(T<=t) one-tail | 0.374 | |
| t Critical one-tail | 1.943 | |
| P(T<=t) two-tail | 0.748 | |
| t Critical two-tail | 2.447 | |

difference in meansbetween the Windows sample and the Mac sample was 75% likely to happen by pure chance (the null hypothesis).
That's way too high a chance, so we say that this data showed **no significant difference** between the Windows and Mac menubars

# Running a T Test

| Win | Mac |
|---|---|
| 625 | 647 |
| 480 | 503 |
| 621 | 559 |
| 633 | 586 |
| 694 | 458 |
| 599 | 380 |
| 505 | 477 |
| 527 | 409 |
| 651 | 589 |
| 505 | 472 |

| | *Windows* | *Mac* |
|---|---|---|
| Mean | 584.0 | 508.1 |
| Variance | 5409.3 | 7295.0 |
| Observations | 10 | 10 |
| Pooled Variance | 6352.2 | |
| Hypothesized Mean Difference | 0 | |
| df | 18 | |
| t Stat | 2.130 | |
| P(T<=t) one-tail | 0.024 | |
| t Critical one-tail | 1.734 | |
| P(T<=t) two-tail | 0.047 | |
| t Critical two-tail | 2.101 | |

The p value for the two-tailed t test is now 0.047, means that the observed difference between the Windows and Mac menubar is only 4.7% likely to happen purely by chance. Assuming we decided to use a 5% significance level, we would now say that the difference is **statistically significant (**$p<0.05$)

## Summary

- Use statistical tests to establish significance of observed differences
- Graphing with error bars is cheap and easy, and great for getting a feel for data
- Use t test to compare two means